

SMALL-DIFFUSIVITY CONCENTRATION FOR FRACTIONAL AGGREGATION WITH CUSP ATTRACTION

RUIYI ZHANG AND XIAOXIN ZHENG

ABSTRACT. We investigate mass concentration induced by cusp attraction in the small-diffusivity regime of the fractional aggregation equation

$$\partial_t u + \varepsilon(-\Delta)^{\gamma/2} u + \nabla \cdot (u \nabla K * u) = 0, \quad K(x) = -|x|, \quad 1 < \gamma < 2.$$

Balancing fractional diffusion with cusp attraction determines the intrinsic scale $R_\varepsilon = \varepsilon^{1/(\gamma-1)}$. For nonnegative radial data, we use the compactly supported Lipschitz inner-mass tent moment

$$\bar{m}_\ell(t) = \int_{\mathbb{R}^N} \left(1 - \frac{|x|}{\ell}\right)_+ u(x, t) \, dx.$$

We prove that the cross-boundary contribution in the fractional pairing is absolutely convergent and nonnegative; together with the quadratic aggregation estimate supplied by radial geometry, this yields the Riccati-type inequality

$$\ell \bar{m}'_\ell(t) \geq \frac{1}{2} \bar{m}_\ell(t)^2 - \varepsilon \mathcal{D}_{\gamma, \ell}(u(t)), \quad \mathcal{D}_{\gamma, R}(u) = a_{N, \gamma} \int_{B_R} \frac{u(x)}{|x|^{\gamma-1}} \, dx.$$

Compared with the classical-diffusion method of Biler–Boritshev–Karch–Laurençot, the fractional operator produces a nonlocal interaction across the boundary of the truncated weight, while the cusp-adapted tent moment leads to a quadratic rather than linear differential inequality for the cusp kernel and radial solutions. Under explicit conditions on the observation time and the dilation factor λ , we prove that the time-integrated mass in $B_{\lambda R_\varepsilon}$ has a positive lower bound independent of ε . Consequently, for every sufficiently small ε , there is a time at which a fixed positive amount of mass lies in a ball of radius comparable to R_ε .

1. INTRODUCTION

Aggregation–diffusion models arise in chemotaxis, collective motion, biological swarms, self-gravitating particle systems, and granular media; see, for example, [1, 2, 28]. In reduced nonlocal chemotaxis models, the convolution field $\nabla K * u$ represents the attraction-induced drift generated by the population density. The connection between nonlocal aggregation equations and Patlak–Keller–Segel models was developed in [3]. In a one-dimensional nonlocal chemotaxis model, Karch–Suzuki [4] studied transient spike formation for attractive kernels. Radial blow-up criteria and structure formation in chemotaxis systems were investigated in [5, 6].

Classical well-posedness, finite-time blow-up, and structural assumptions on the interaction potential have been studied in [7, 8, 9]. Fractionally dissipative aggregation equations were considered by Li–Rodrigo [21, 22] and Biler–Karch–Laurençot [10]; related fractional Keller–Segel models appear in [11, 23, 24]. Diffusion–aggregation equations with bounded or irregular interaction kernels were studied in [25, 26].

This work builds on the analysis by Biler–Boritshev–Karch–Laurençot [12] of small-diffusivity mass concentration for classical aggregation–diffusion equations. Their argument uses a regularized truncated-distance moment. In their moment inequality, the diffusion term produces the observable $u(0, t)$ in one space dimension and $\int_{\mathbb{R}^N} u(x, t)/|x| \, dx$ in higher dimensions, leading to an ε -uniform mass lower bound near the origin on the scale $R_\varepsilon \sim \varepsilon$. Subsequent work [13] quantified the growth of Sobolev norms for the model kernel $K = -|x|$.

For radial initial data, we study the fractional counterpart

$$(1) \quad \partial_t u + \varepsilon \Lambda^\gamma u + \nabla \cdot (u \nabla K * u) = 0, \quad K(x) = -|x|, \quad 1 < \gamma < 2,$$

where $\varepsilon > 0$ and $\Lambda^\gamma = (-\Delta)^{\gamma/2}$. On a spatial scale R , fractional diffusion has strength of order $\varepsilon R^{-\gamma}$, whereas cusp attraction has compressive strength of order R^{-1} . Balancing these effects yields the

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intrinsic scale

$$R_\varepsilon = \varepsilon^{1/(\gamma-1)}.$$

Our aim is to obtain a positive lower bound, uniform in ε , for the time-integrated mass in balls of radius λR_ε . The result holds provided the fixed observation time and dilation factor λ satisfy explicit lower bounds. The bound $|\nabla K| \leq 1$ also permits a direct construction of the global mild solutions used in the concentration argument.

The difference from the classical-diffusion argument is already visible in the diffusion term. Biler–Boritschev–Karch–Laurençot [12] use a regularized truncated-distance moment, while the corresponding local structure can be illustrated by the distributional identity for the unsmoothed tent:

$$(-\Delta) \left(1 - \frac{|x|}{\ell}\right)_+ = \begin{cases} \frac{N-1}{\ell|x|} \mathbf{1}_{B_\ell} - \frac{1}{\ell} \mathcal{H}^{N-1}|_{\partial B_\ell}, & N \geq 2, \\ \frac{2}{\ell} \delta_0 - \frac{1}{\ell} (\delta_{-\ell} + \delta_\ell), & N = 1. \end{cases}$$

Thus local diffusion produces an interior contribution when $N \geq 2$ and a point mass at the origin when $N = 1$; in both cases, a boundary measure is also present. By contrast, $\Lambda^\gamma(1 - |x|/\ell)_+$ does not localize on ∂B_ℓ : its pairing with u couples B_ℓ to $\mathbb{R}^N \setminus B_\ell$. We prove that this cross-boundary contribution is absolutely convergent and nonnegative.

The choice of the tent weight is guided by two complementary observations. First,

$$\left(1 - \frac{|x|}{\ell}\right)_+ = \frac{1}{\ell} \int_0^\ell \mathbf{1}_{B_r}(x) \, dr.$$

Consequently, we have

$$m_r(t) := \int_{B_r} u(x, t) \, dx, \quad \bar{m}_\ell(t) := \int_{\mathbb{R}^N} \left(1 - \frac{|x|}{\ell}\right)_+ u(x, t) \, dx = \frac{1}{\ell} \int_0^\ell m_r(t) \, dr.$$

Thus $\bar{m}_\ell$ is the scale average of the mass in balls centered at the origin.

Second, the inward radial gradient

$$\nabla \left(1 - \frac{|x|}{\ell}\right)_+ = -\frac{1}{\ell} \frac{x}{|x|} \mathbf{1}_{B_\ell}(x)$$

interacts favorably with the attractive cusp force in the symmetrized aggregation pairing. Indeed,

$$\nabla K(x - y) = -\frac{x - y}{|x - y|},$$

and for radial solutions, symmetrization and angular averaging give

$$\int_{\mathbb{R}^N} u(x, t) (\nabla K * u(t))(x) \cdot \nabla \left(1 - \frac{|x|}{\ell}\right)_+ \, dx \geq \frac{1}{2\ell} m_\ell(t)^2.$$

Thus the same weight has both a direct mass interpretation and the geometric structure needed for the Riccati argument. The function $(1 - |x|/\ell)_+$ belongs to $W^{1,\infty}(\mathbb{R}^N)$ and is supported in $\bar{B}_\ell$. Hence $(1 - |x|/\ell)_+ \in H^1(\mathbb{R}^N) \subset H^{\gamma/2}(\mathbb{R}^N)$.

For $R > 0$, define the weighted diffusion quantity

$$\mathcal{D}_{\gamma,R}(u) := a_{N,\gamma} \int_{B_R} \frac{u(x)}{|x|^{\gamma-1}} \, dx.$$

Fractional diffusion acts on the same weight. We prove

$$\left\langle \Lambda^\gamma u, \left(1 - \frac{|x|}{\ell}\right)_+ \right\rangle = \frac{1}{\ell} \mathcal{D}_{\gamma,\ell}(u) - c_{N,\gamma} \int_{B_\ell} \int_{\mathbb{R}^N \setminus B_\ell} \frac{u(x) \left(\frac{|y|}{\ell} - 1\right) + u(y) \left(1 - \frac{|x|}{\ell}\right)}{|x - y|^{N+\gamma}} \, dy \, dx.$$

Here $a_{N,\gamma} > 0$ and $c_{N,\gamma} > 0$ are specified in Subsection 3.1. The cross-boundary integral is nonnegative. Combining this decomposition with the radial aggregation bound gives

$$\ell \bar{m}'_\ell(t) \geq \frac{1}{2} \bar{m}_\ell(t)^2 - \varepsilon \mathcal{D}_{\gamma,\ell}(u(t)).$$

This Riccati-type inequality leads to a time-integrated lower bound for $\mathcal{D}_{\gamma,\ell}(u)$.

Remark 1.1 (Comparison with the method of Biler–Borichev–Karch–Laurençot). The present argument differs from that of [12] in both the choice of moment and the resulting differential inequality. They use a regularized increasing truncated-distance weight, whereas we use the decreasing inner-mass tent $(1 - |x|/\ell)_+$ without smoothing. The $W^{1,\infty} \cap H^{\gamma/2}$ regularity stated above is sufficient for the first-order aggregation pairing and the fractional weak formulation. The contribution of the kink across ∂B_ℓ is handled directly through the nonlocal cross-boundary decomposition.

The resulting moment dynamics are also different. The truncated-distance moment in [12] satisfies a differential inequality that is linear in the evolving moment, with an exterior-interaction error handled by an integrating factor. For the cusp kernel $K(x) = -|x|$, radial angular averaging makes the mixed inside–outside interactions nonnegative and yields a quadratic lower bound from the interaction inside B_ℓ . This gives the Riccati-type inequality above. In one dimension, the locally integrable weighted observable replaces the point value $u(0, t)$; see Remark 4.8. The Riccati structure relies on the specific geometry of the cusp kernel, while [12] applies to a broader class of radial attractive kernels.

The remaining step is to pass from the weighted estimate to mass in a ball. Since $|x|^{-(\gamma-1)}$ belongs to weak $L^{N/(\gamma-1)}$, a Lorentz-space inequality yields this conversion once a scale-compatible finite- p bound is available. The equation supplies the required bound. Applying the conversion at $R = \lambda R_\varepsilon$ yields the desired time-integrated mass lower bound.

The present argument relies on radial symmetry and the geometry of the cusp kernel, and extending it to more general kernels or nonradial solutions would require new coercivity estimates. The choice of the linear tent also suggests a possible extension. For $0 < \beta \leq 1$, set $\psi_{\beta,\ell}(x) := [1 - (|x|/\ell)^\beta]_+$. For exponents satisfying $\gamma - \beta < N$, the Riesz-potential identity $\Lambda^\gamma |x|^\beta = -a_{N,\gamma,\beta} |x|^{\beta-\gamma}$, $x \neq 0$, with $a_{N,\gamma,\beta} > 0$, suggests the candidate scale $R_{\varepsilon,\beta} \sim \varepsilon^{1/(\gamma-\beta)}$ at the level of the diffusion estimate. For $0 < \beta < 1$, however, $|\nabla \psi_{\beta,\ell}(x)| = \beta \ell^{-\beta} |x|^{\beta-1}$ for $0 < |x| < \ell$, so the gradient is singular at the origin. A rigorous extension would therefore require both an admissibility result for this less regular test function and a weighted angular coercivity estimate for the aggregation term. Within this formal family, $\beta = 1$ is the Lipschitz endpoint and, for $0 < \varepsilon < 1$, yields the smallest candidate ε -scale.

The paper is organized as follows. Section 2 states the well-posedness and concentration results. Section 3 develops the weighted concentration mechanism, including the fractional diffusion decomposition, the radial aggregation bound, and the Riccati alternative. Section 4 converts the weighted estimate into small-ball concentration at the intrinsic scale and proves the final concentration results. Appendix A contains the mild-solution framework and the passage to the tent weight in the weak formulation.

2. MAIN RESULTS

We begin with the notation and solution framework used throughout, and then state the concentration theorem and its direct consequences.

2.1. Notation and conventions. We write $B_R := \{x \in \mathbb{R}^N : |x| < R\}$, $p' := p/(p-1)$, and $s_+ := \max\{s, 0\}$, $s_- := \max\{-s, 0\}$. The operator Λ^γ denotes $(-\Delta)^{\gamma/2}$, with the normalization specified in Subsection 3.1. We write $\sigma_N := |\mathbb{S}^{N-1}|$ for the surface area of the unit sphere in $\mathbb{R}^N$, and $\int_{\mathbb{S}^{N-1}}$ denotes integration with respect to normalized surface measure. The symbols C and c , possibly with parameter subscripts, denote positive generic constants whose values may change from line to line; their parameter dependence is indicated when relevant. Constants with descriptive subscripts or superscripts are fixed within the estimates where they are introduced. The fractional-normalization constant $c_{N,\gamma}$ and the structural constant $a_{N,\gamma}$ are tracked explicitly and are not used as generic constants. Unless stated otherwise, spatial integrals are taken over $\mathbb{R}^N$. For the cusp kernel, we set $\nabla K(0) = 0$. Throughout, $N \geq 1$ and $1 < \gamma < 2$.

Theorem 2.1 (Global mild well-posedness). *Let $u_0 \geq 0$ satisfy*

$$u_0 \in L^1(\mathbb{R}^N) \cap L^\infty(\mathbb{R}^N), \quad M := \int_{\mathbb{R}^N} u_0(x) \, dx > 0.$$

Fix any

$$p_* > \max \left\{ 2, \frac{N}{\gamma-1} \right\}.$$

For every $\varepsilon > 0$, equation (1) admits a global mild solution

$$u \in C([0, \infty); L^1(\mathbb{R}^N) \cap L^{p^*}(\mathbb{R}^N)) \cap L_{\text{loc}}^\infty([0, \infty); L^\infty(\mathbb{R}^N))$$

satisfying

$$(2) \quad u(t) = e^{-\varepsilon t \Lambda^\gamma} u_0 - \int_0^t \nabla \cdot e^{-\varepsilon(t-s)\Lambda^\gamma} [u(s)(\nabla K * u(s))] \, ds.$$

For every $T > 0$, this solution is unique among functions in

$$C([0, T]; L^1(\mathbb{R}^N) \cap L^{p^*}(\mathbb{R}^N)) \cap L^\infty(0, T; L^\infty(\mathbb{R}^N))$$

that satisfy (2) on $[0, T]$. The solution is nonnegative and conserves mass:

$$\int_{\mathbb{R}^N} u(x, t) \, dx = M, \quad t \geq 0.$$

For every $\tau > 0$, the positive-time bound

$$u \in L^\infty([\tau, \infty); L^\infty(\mathbb{R}^N))$$

holds, and for every finite $p \geq 2$, the finite- p energy inequality of Subsection 4.2 holds. If u_0 is radial, then $u(\cdot, t)$ remains radial. Moreover, for every $\ell > 0$, the tent moment $\bar{m}_\ell$ belongs to $AC_{\text{loc}}([0, \infty))$, and the integrated tent-moment identity (31) holds.

Remark 2.2. The well-posedness theorem applies to general initial data. The concentration results assume radial symmetry in order to use the angular-average lower bound for the aggregation term. Appendix A proves Theorem 2.1, with global continuation supplied by the finite- p estimate of Subsection 4.2.

In the small-diffusivity limit, the initial datum is fixed independently of ε . We omit the superscript ε when it is clear from the context. All constants, unless otherwise specified, depend only on the parameters explicitly indicated and are independent of ε .

2.2. Main concentration result. We now state the concentration theorem and its direct consequences.

Theorem 2.3 (Time-integrated mass concentration at the intrinsic scale in the small-diffusivity regime). *Let $p \in [2, \infty)$ and*

$$p > \frac{N}{N+1-\gamma}.$$

Let $u_0 \geq 0$ be radial and satisfy $u_0 \in L^1(\mathbb{R}^N) \cap L^\infty(\mathbb{R}^N)$, with total mass $M > 0$. Let $\ell > 0$ and set

$$\bar{m}_\ell(0) := \int_{\mathbb{R}^N} \left(1 - \frac{|x|}{\ell}\right)_+ u_0(x) \, dx > 0,$$

and assume

$$T \geq \frac{6\ell}{\bar{m}_\ell(0)} - \frac{2\ell}{M}.$$

For each $\varepsilon > 0$, let u^ε be the unique global mild solution given by Theorem 2.1, which remains radial for every $t \geq 0$. If $\lambda > 0$ satisfies

$$\lambda^{\gamma-1} > \frac{2a_{N,\gamma}TM}{\ell\bar{m}_\ell(0)},$$

then there exist $\varepsilon_ > 0$ and $C_* > 0$ such that for all $0 < \varepsilon < \varepsilon_*$,*

$$(3) \quad \int_0^T \int_{B_{\lambda\varepsilon^{1/(\gamma-1)}}} u^\varepsilon(x, t) \, dx \, dt \geq C_*.$$

The constants ε_ and C_* may depend on*

$$N, \gamma, p, M, \|u_0\|_{L^p}, \ell, \bar{m}_\ell(0), T, \lambda$$

but are independent of ε . Thus the time-integrated mass over $[0, T]$ in balls of radius λR_ε remains bounded away from zero uniformly as $\varepsilon \rightarrow 0$, where $R_\varepsilon = \varepsilon^{1/(\gamma-1)}$. The threshold ε_ may be chosen so that $\lambda\varepsilon^{1/(\gamma-1)} < \ell$ for every $0 < \varepsilon < \varepsilon_*$.*

Corollary 2.4 (Concentration at some time). *Under the assumptions of Theorem 2.3, for every $0 < \varepsilon < \varepsilon_*$ there exists a time $t_\varepsilon \in [0, T]$ such that*

$$(4) \quad \int_{B_{\lambda\varepsilon^{1/(\gamma-1)}}} u^\varepsilon(x, t_\varepsilon) dx \geq \frac{C_*}{T}.$$

Thus, for each sufficiently small diffusivity, there is a time at which the mass in a ball of radius λR_ε is bounded below independently of ε .

Remark 2.5. The Lorentz-space conversion uses

$$p > \frac{N}{N+1-\gamma}, \quad 0 < \frac{(\gamma-1)p'}{N} < 1.$$

The energy estimate in Proposition 4.5 additionally uses $p \geq 2$.

Corollary 2.6 (Time-integrated lower bound for local L^q norms on small balls). *Under the assumptions of Theorem 2.3, for every $1 \leq q < \infty$, there exists a constant $C_q > 0$, independent of ε , such that for all $0 < \varepsilon < \varepsilon_*$,*

$$(5) \quad \int_0^T \left(\int_{B_{\lambda\varepsilon^{1/(\gamma-1)}}} u^\varepsilon(x, t)^q dx \right)^{1/q} dt \geq C_q \varepsilon^{-\frac{N(q-1)}{q(\gamma-1)}}.$$

3. WEIGHTED CONCENTRATION MECHANISM

The concentration mechanism has three steps. We first decompose the fractional pairing with the tent weight, then derive a radial lower bound for the aggregation term, and finally combine these estimates in a Riccati alternative. This chain produces the weighted lower bound that will be converted into small-ball mass in Section 4.

3.1. Decomposition of the fractional diffusion contribution. Set $\phi_\ell(x) := (1 - |x|/\ell)_+$. Proposition A.5 shows that it is admissible in the weak formulation of the global mild solution and that the resulting pairing is integrable in time. We normalize Λ^γ by $\widehat{\Lambda^\gamma f}(\xi) = |\xi|^\gamma \widehat{f}(\xi)$. Equivalently,

$$\Lambda^\gamma f(x) = c_{N,\gamma} \text{p. v.} \int_{\mathbb{R}^N} \frac{f(x) - f(y)}{|x - y|^{N+\gamma}} dy, \quad c_{N,\gamma} = \frac{2^\gamma \Gamma\left(\frac{N+\gamma}{2}\right)}{\pi^{N/2} |\Gamma(-\gamma/2)|} > 0.$$

With this normalization, the standard Riesz-potential identity [20] gives

$$\Lambda^\gamma |x| = -a_{N,\gamma} |x|^{1-\gamma}, \quad x \neq 0, \quad a_{N,\gamma} = 2^{\gamma-1} \frac{\Gamma\left(\frac{N+1}{2}\right) \Gamma\left(\frac{\gamma-1}{2}\right)}{\sqrt{\pi} \Gamma\left(\frac{N+1-\gamma}{2}\right)} > 0.$$

The pairing in Proposition 3.1 is understood in the transposition sense; its defining integral is proved below to be absolutely convergent. Proposition A.5 justifies its agreement with the Dirichlet pairing for the mild solution at almost every positive time.

Proposition 3.1 (Fractional diffusion identity for the tent weight). *Let $\ell > 0$, $u \geq 0$, and $u \in L^1(\mathbb{R}^N) \cap L^\infty(\mathbb{R}^N)$. Then*

$$(6) \quad \left\langle \Lambda^\gamma u, \left(1 - \frac{|x|}{\ell}\right)_+ \right\rangle := \int_{\mathbb{R}^N} u(x) \Lambda^\gamma \left(1 - \frac{|x|}{\ell}\right)_+ dx \\ = \frac{1}{\ell} \mathcal{D}_{\gamma,\ell}(u) - c_{N,\gamma} \int_{B_\ell} \int_{\mathbb{R}^N \setminus B_\ell} \frac{u(x) \left(\frac{|y|}{\ell} - 1\right) + u(y) \left(1 - \frac{|x|}{\ell}\right)}{|x - y|^{N+\gamma}} dy dx.$$

The double integral is nonnegative. Hence

$$\left\langle \Lambda^\gamma u, \left(1 - \frac{|x|}{\ell}\right)_+ \right\rangle \leq \frac{1}{\ell} \mathcal{D}_{\gamma,\ell}(u).$$

Proof. For $0 < |x| < \ell$,

$$\left(1 - \frac{|x|}{\ell}\right)_+ = 1 - \frac{|x|}{\ell} + \frac{(|x| - \ell)_+}{\ell},$$

and therefore

$$(7) \quad \Lambda^\gamma \left(1 - \frac{|x|}{\ell}\right)_+ = \frac{a_{N,\gamma}}{\ell|x|^{\gamma-1}} - \frac{c_{N,\gamma}}{\ell} \int_{\mathbb{R}^N \setminus B_\ell} \frac{|y| - \ell}{|x - y|^{N+\gamma}} dy.$$

For $|x| > \ell$, the tent weight vanishes and

$$(8) \quad \Lambda^\gamma \left(1 - \frac{|x|}{\ell}\right)_+ = -c_{N,\gamma} \int_{B_\ell} \frac{1 - |y|/\ell}{|x - y|^{N+\gamma}} dy.$$

The interface is a null set, so it remains to justify absolute integrability of the cross-boundary terms. For $x \in B_\ell$ and $y \notin B_\ell$, the triangle inequality gives $\ell - |x| \leq |x - y|$ and $|y| - \ell \leq |x - y|$. Near ∂B_ℓ , the numerator of the cross-boundary kernel is therefore bounded by

$$u(x) \left(\frac{|y|}{\ell} - 1\right) + u(y) \left(1 - \frac{|x|}{\ell}\right) \leq \frac{\|u\|_{L^\infty}}{\ell} ((|y| - \ell) + (\ell - |x|)).$$

We estimate the possible boundary singularity directly in spherical coordinates. Fix $\delta \in (0, \min\{\ell/2, 1\})$. Since the kernel is nonsingular away from the interface, it is enough to consider $\ell - \delta < |x| < \ell$ and $\ell < |y| < \ell + \delta$. Write

$$x = (\ell - s)\omega, \quad y = (\ell + r)\sigma, \quad 0 < s, r < \delta, \quad \omega, \sigma \in \mathbb{S}^{N-1}.$$

Then $s = \ell - |x|$ and $r = |y| - \ell$. A direct computation gives the identity

$$(9) \quad |x - y|^2 = |(\ell - s)\omega - (\ell + r)\sigma|^2 = (s + r)^2 + (\ell - s)(\ell + r)|\omega - \sigma|^2.$$

Indeed, $|\omega - \sigma|^2 = 2 - 2\omega \cdot \sigma$. Since $0 < s, r < \delta < \ell/2$, $\ell^2/2 \leq (\ell - s)(\ell + r) \leq 3\ell^2/2$. In particular, (9) implies the lower bound

$$|x - y|^2 \geq (s + r)^2 + \frac{\ell^2}{2} |\omega - \sigma|^2.$$

The Jacobian in these variables is

$$dx dy = (\ell - s)^{N-1} (\ell + r)^{N-1} ds dr d\omega d\sigma, \quad (\ell - s)^{N-1} (\ell + r)^{N-1} \leq \left(\frac{3\ell^2}{2}\right)^{N-1}.$$

Combining this estimate with the numerator bound above, the contribution from the boundary layer satisfies

$$(10) \quad \begin{aligned} & \int_{\ell-\delta < |x| < \ell} \int_{\ell < |y| < \ell+\delta} \frac{u(x)(|y|/\ell - 1) + u(y)(1 - |x|/\ell)}{|x - y|^{N+\gamma}} dy dx \\ & \leq \frac{\|u\|_{L^\infty}}{\ell} \left(\frac{3\ell^2}{2}\right)^{N-1} \int_0^\delta \int_0^\delta \int_{\mathbb{S}^{N-1}} \int_{\mathbb{S}^{N-1}} \frac{s + r}{\left((s + r)^2 + \frac{\ell^2}{2} |\omega - \sigma|^2\right)^{\frac{N+\gamma}{2}}} d\sigma d\omega dr ds. \end{aligned}$$

Assume first that $N \geq 2$, and fix $\omega \in \mathbb{S}^{N-1}$. Let $\theta \in [0, \pi]$ denote the angle between ω and σ . Since $|\omega - \sigma| = 2\sin(\theta/2)$, geodesic polar coordinates centered at ω give

$$\int_{\mathbb{S}^{N-1}} \frac{s + r}{\left((s + r)^2 + \frac{\ell^2}{2} |\omega - \sigma|^2\right)^{\frac{N+\gamma}{2}}} d\sigma = \sigma_{N-1} \int_0^\pi \frac{(s + r) \sin^{N-2} \theta}{\left((s + r)^2 + 2\ell^2 \sin^2(\theta/2)\right)^{\frac{N+\gamma}{2}}} d\theta.$$

For $0 \leq \theta \leq \pi$, we have $\sin \theta \leq \theta$ and $\sin(\theta/2) \geq \theta/\pi$. Therefore, after the change of variables $\rho = \ell\theta/(s + r)$,

$$\int_{\mathbb{S}^{N-1}} \frac{(s + r) d\sigma}{\left((s + r)^2 + \frac{\ell^2}{2} |\omega - \sigma|^2\right)^{\frac{N+\gamma}{2}}} \leq \frac{\sigma_{N-1}}{\ell^{N-1}} (s + r)^{-\gamma} \int_0^{\frac{\pi\ell}{s+r}} \frac{\rho^{N-2} d\rho}{\left(1 + \frac{2}{\pi^2} \rho^2\right)^{\frac{N+\gamma}{2}}} \lesssim_{N,\gamma,\ell} (s + r)^{-\gamma}.$$

The implicit constant is independent of s, r, δ , and u . Using this estimate in (10) and integrating in ω , it remains to note that

$$\int_0^\delta \int_0^\delta (s + r)^{-\gamma} dr ds = \frac{2 - 2^{2-\gamma}}{(\gamma - 1)(2 - \gamma)} \delta^{2-\gamma} < \infty$$

because $\gamma < 2$.

When $N = 1$, the boundary consists of the two points $\pm\ell$. Near either boundary point one may write $|x - y| = s + r$, and the corresponding estimate is simply

$$\frac{\|u\|_{L^\infty}}{\ell} \int_0^\delta \int_0^\delta \frac{s+r}{(s+r)^{1+\gamma}} dr ds = \frac{\|u\|_{L^\infty}}{\ell} \frac{2-2^{2-\gamma}}{(\gamma-1)(2-\gamma)} \delta^{2-\gamma} < \infty.$$

The two same-side boundary contributions are therefore finite; the opposite-side contributions are nonsingular. Away from the interface the kernel is nonsingular. At infinity, the term containing $u(x)(|y|/\ell - 1)$ decays like $|y|^{-N-\gamma+1}$, which is integrable because $\gamma > 1$, while the term containing $u(y)(1 - |x|/\ell)$ is controlled by $u(y)|y|^{-N-\gamma}$ and is integrable since $u \in L^1$. Thus the two cross-boundary terms are absolutely integrable. Fubini's theorem and the exchange of the interior and exterior variables are justified; multiplying (7) and (8) by u , integrating, and exchanging x and y in the exterior contribution gives (6). $\square$

3.2. Aggregation growth of the truncated inner-mass moment. Since

$$\nabla \left(1 - \frac{|x|}{\ell} \right)_+ = -\frac{1}{\ell} \frac{x}{|x|} \mathbf{1}_{B_\ell}(x) \quad \text{for a.e. } x \neq 0,$$

the aggregation contribution is localized to B_ℓ .

Lemma 3.2 (Angular-average geometric lower bound). *Let $N \geq 2$ and $\ell > 0$. For a.e. $r, s > 0$,*

$$(11) \quad \oint_{\mathbb{S}^{N-1}} \oint_{\mathbb{S}^{N-1}} \frac{r\omega - s\sigma}{|r\omega - s\sigma|} \cdot (\mathbf{1}_{\{r < \ell\}}\omega - \mathbf{1}_{\{s < \ell\}}\sigma) d\omega d\sigma \geq \mathbf{1}_{\{r < \ell\}} \mathbf{1}_{\{s < \ell\}}.$$

Proof. The notation $\oint_{\mathbb{S}^{N-1}}$ denotes the normalized spherical average. For $b \geq 0$, fix $\omega_0 \in \mathbb{S}^{N-1}$ and define

$$h_b(r) := \oint_{\mathbb{S}^{N-1}} |r\omega_0 - b\sigma| d\sigma, \quad r \geq 0.$$

By rotation invariance, h_b is independent of the choice of ω_0 . To see that h_b is nondecreasing, note first that $x \mapsto \oint_{\mathbb{S}^{N-1}} |x - b\sigma| d\sigma$ is convex, since it is an average of convex functions. It is also radial by rotation invariance. Its radial profile is precisely h_b , and a radial convex function attains its minimum at the origin. Hence h_b is nondecreasing on $[0, \infty)$, so $h'_b(r) \geq 0$ at every differentiability point.

At such points, dominated convergence gives

$$h'_s(r) = \oint_{\mathbb{S}^{N-1}} \frac{r - s(\omega_0 \cdot \sigma)}{|r\omega_0 - s\sigma|} d\sigma.$$

By rotation invariance, the part of the double average in (11) containing ω equals $\mathbf{1}_{\{r < \ell\}} h'_s(r)$, and the part containing σ equals $\mathbf{1}_{\{s < \ell\}} h'_r(s)$. Thus the whole double average is

$$\oint_{\mathbb{S}^{N-1}} \oint_{\mathbb{S}^{N-1}} \frac{r\omega - s\sigma}{|r\omega - s\sigma|} \cdot (\mathbf{1}_{\{r < \ell\}}\omega - \mathbf{1}_{\{s < \ell\}}\sigma) d\omega d\sigma = \mathbf{1}_{\{r < \ell\}} h'_s(r) + \mathbf{1}_{\{s < \ell\}} h'_r(s).$$

If exactly one of r, s lies in $(0, \ell)$, this quantity is nonnegative by the monotonicity just proved, while the right-hand side of (11) is zero. If both radii are at least ℓ , both sides vanish.

It remains to consider $r, s < \ell$. In this case the left-hand side of (11) is

$$\oint_{\mathbb{S}^{N-1}} \oint_{\mathbb{S}^{N-1}} \frac{r\omega - s\sigma}{|r\omega - s\sigma|} \cdot (\omega - \sigma) d\omega d\sigma = \oint_{\mathbb{S}^{N-1}} \oint_{\mathbb{S}^{N-1}} \frac{(r+s)(1 - \omega \cdot \sigma)}{|r\omega - s\sigma|} d\omega d\sigma.$$

Since $|r\omega - s\sigma| \leq r + s$, we have the pointwise lower bound

$$\frac{(r+s)(1 - \omega \cdot \sigma)}{|r\omega - s\sigma|} \geq 1 - \omega \cdot \sigma.$$

Taking the double normalized spherical average and using

$$\oint_{\mathbb{S}^{N-1}} \oint_{\mathbb{S}^{N-1}} \omega \cdot \sigma d\omega d\sigma = \left(\oint_{\mathbb{S}^{N-1}} \omega d\omega \right) \cdot \left(\oint_{\mathbb{S}^{N-1}} \sigma d\sigma \right) = 0,$$

we obtain the lower bound 1. This proves (11). $\square$

Proposition 3.3 (Radial-average growth of the truncated inner-mass moment). *Let $\ell > 0$, let $K(x) = -|x|$, and let $u(\cdot, t) \geq 0$ be radial. Then the aggregation contribution in the evolution of $\bar{m}_\ell$ satisfies*

$$(12) \quad \frac{1}{2\ell} \iint_{\mathbb{R}^N \times \mathbb{R}^N} u(x, t) u(y, t) \frac{x - y}{|x - y|} \cdot \left(\mathbf{1}_{B_\ell}(x) \frac{x}{|x|} - \mathbf{1}_{B_\ell}(y) \frac{y}{|y|} \right) dx dy \geq \frac{1}{2\ell} m_\ell(t)^2,$$

where $m_\ell(t) := \int_{B_\ell} u(x, t) dx$.

Proof. We omit the time variable. All identities below are understood almost everywhere; the sets on which $x = y$, $x = 0$, or $y = 0$ have measure zero.

For $N = 1$, evenness gives, for $x > 0$,

$$(K' * u)(x) = -2 \int_0^x u(y) dy.$$

Together with the almost-everywhere identity

$$\frac{d}{dx} \left(1 - \frac{|x|}{\ell} \right)_+ = -\frac{1}{\ell} \frac{x}{|x|} \mathbf{1}_{B_\ell}(x),$$

the aggregation contribution equals

$$\frac{4}{\ell} \int_0^\ell u(x) \int_0^x u(y) dy dx = \frac{1}{2\ell} m_\ell^2.$$

For $N \geq 2$, symmetrization and polar coordinates give the following representation of the left-hand side of (12):

$$\frac{1}{2\ell} \int_0^\infty \int_0^\infty u(r) u(s) r^{N-1} s^{N-1} \int_{\mathbb{S}^{N-1}} \int_{\mathbb{S}^{N-1}} \frac{r\omega - s\sigma}{|r\omega - s\sigma|} \cdot (\mathbf{1}_{\{r < \ell\}} \omega - \mathbf{1}_{\{s < \ell\}} \sigma) d\omega d\sigma dr ds.$$

Since each ordinary spherical integral is σ_N times the corresponding normalized average, Lemma 3.2 gives a lower bound $\sigma_N^2 \mathbf{1}_{\{r < \ell\}} \mathbf{1}_{\{s < \ell\}}$ for the angular integral. Hence the left-hand side of (12) is bounded below by

$$\frac{1}{2\ell} \sigma_N^2 \left(\int_0^\ell u(r) r^{N-1} dr \right)^2 = \frac{1}{2\ell} m_\ell^2.$$

□

3.3. The Riccati alternative. From the definition,

$$0 \leq \bar{m}_\ell(t) \leq m_\ell(t) \leq M.$$

Theorem 2.1 and Proposition A.5 imply that $\bar{m}_\ell \in AC([0, T])$ for every $T > 0$; hence the hitting times used below are well defined.

Proposition 3.4 (Riccati moment inequality). *Let u be the nonnegative radial global mild solution of (1) with $K = -|x|$, diffusivity $\varepsilon > 0$, and conserved mass M . Then, for every $0 \leq t_1 \leq t_2 < \infty$,*

$$(13) \quad \ell(\bar{m}_\ell(t_2) - \bar{m}_\ell(t_1)) \geq \frac{1}{2} \int_{t_1}^{t_2} \bar{m}_\ell(t)^2 dt - \varepsilon \int_{t_1}^{t_2} \mathcal{D}_{\gamma, \ell}(u(t)) dt.$$

For almost every time,

$$(14) \quad \ell \bar{m}'_\ell(t) \geq \frac{1}{2} \bar{m}_\ell(t)^2 - \varepsilon \mathcal{D}_{\gamma, \ell}(u(t)).$$

Proof. Differentiating the integrated tent-moment identity (31) from Proposition A.5 at almost every time and then symmetrizing the aggregation term gives

$$\begin{aligned} \frac{d}{dt} \bar{m}_\ell(t) &= -\varepsilon \left\langle \Lambda^\gamma u(t), \left(1 - \frac{|x|}{\ell} \right)_+ \right\rangle \\ &\quad + \frac{1}{2\ell} \iint_{\mathbb{R}^N \times \mathbb{R}^N} u(x, t) u(y, t) \frac{x - y}{|x - y|} \cdot \left(\mathbf{1}_{B_\ell}(x) \frac{x}{|x|} - \mathbf{1}_{B_\ell}(y) \frac{y}{|y|} \right) dx dy. \end{aligned}$$

Proposition 3.1 and Proposition 3.3 yield

$$\ell \bar{m}'_\ell(t) \geq \frac{1}{2} m_\ell(t)^2 - \varepsilon \mathcal{D}_{\gamma, \ell}(u(t)).$$

Since $m_\ell(t) \geq \bar{m}_\ell(t)$, this proves (14); integration gives (13). □

Theorem 3.5 (Riccati-type weighted concentration). *Let u be the nonnegative radial global mild solution of (1) with $K = -|x|$, diffusivity $\varepsilon > 0$, and conserved mass M . Let $\ell > 0$ and assume $\bar{m}_\ell(0) > 0$. If*

$$T \geq \frac{6\ell}{\bar{m}_\ell(0)} - \frac{2\ell}{M},$$

then

$$(15) \quad \int_0^T \mathcal{D}_{\gamma,\ell}(u(t)) \, dt \geq \frac{\ell \bar{m}_\ell(0)}{2\varepsilon}.$$

Proof. We use the absolutely continuous representative of $\bar{m}_\ell$. If $\bar{m}_\ell$ reaches $\bar{m}_\ell(0)/2$ on $[0, T]$, let τ be the first such time. Integrating (14) on $[0, \tau]$ and discarding the quadratic term gives

$$\varepsilon \int_0^\tau \mathcal{D}_{\gamma,\ell}(u(t)) \, dt \geq \ell (\bar{m}_\ell(0) - \bar{m}_\ell(\tau)) = \frac{\ell \bar{m}_\ell(0)}{2}.$$

Assume next that

$$\bar{m}_\ell(t) > \frac{1}{2} \bar{m}_\ell(0), \quad 0 \leq t \leq T.$$

Dividing (14) by $\ell \bar{m}_\ell(t)^2$ gives

$$\left(\frac{1}{\bar{m}_\ell(t)} \right)' \leq -\frac{1}{2\ell} + \frac{4\varepsilon \mathcal{D}_{\gamma,\ell}(u(t))}{\ell \bar{m}_\ell(0)^2}.$$

After integration and the bound $\bar{m}_\ell(T) \leq M$,

$$\varepsilon \int_0^T \mathcal{D}_{\gamma,\ell}(u(t)) \, dt \geq \frac{\bar{m}_\ell(0)^2}{4} \left[\frac{T}{2} - \ell \left(\frac{1}{\bar{m}_\ell(0)} - \frac{1}{M} \right) \right].$$

The assumed lower bound on T makes the bracket at least $2\ell/\bar{m}_\ell(0)$, which again yields (15). $\square$

Remark 3.6 (Existence of a scale with positive initial tent moment). For every nontrivial nonnegative finite-mass datum u_0 , there is a finite scale $\ell > 0$ such that $\bar{m}_\ell(0) > 0$. Indeed, choose $R < \infty$ with $\int_{B_R} u_0(x) \, dx > 0$, and then take $\ell > R$. Since

$$\bar{m}_\ell(0) = \int_{B_\ell} \left(1 - \frac{|x|}{\ell} \right) u_0(x) \, dx \geq \left(1 - \frac{R}{\ell} \right) \int_{B_R} u_0(x) \, dx > 0,$$

the condition $\bar{m}_\ell(0) > 0$ can always be met at a finite scale. For that choice of ℓ , the theorem applies whenever $T \geq 6\ell/\bar{m}_\ell(0) - 2\ell/M$.

4. SMALL-BALL CONCENTRATION AT THE INTRINSIC SCALE

4.1. Weighted-to-unweighted conversion. We use the notation $L^{p,q}$ for Lorentz spaces. If f^* is the nonincreasing rearrangement of f , our conventions are

$$\|f\|_{L^{p,1}} := \int_0^\infty t^{1/p} f^*(t) \frac{dt}{t}, \quad \|f\|_{L^{p,\infty}} := \sup_{t>0} t^{1/p} f^*(t).$$

With this normalization, the endpoint Lorentz–Hölder constant in $L^{p,1} \times L^{p',\infty} \rightarrow L^1$ is one.

Combined with interpolation, the weak-space estimate converts $\mathcal{D}_{\gamma,R}(u)$ into a lower bound for the unweighted small-ball mass $\int_{B_R} u(x) \, dx$. The estimate respects mass-preserving spatial rescaling: $R^{\gamma-1} \mathcal{D}_{\gamma,R}$, $\int_{B_R} u(x) \, dx$, and $R^{N/p'} \|u\|_{L^p(B_R)}$ have the same mass homogeneity.

The weighted estimate of Theorem 3.5 can first be localized to a smaller scale. For $0 < R < \ell$,

$$\mathcal{D}_{\gamma,\ell}(u) \leq \mathcal{D}_{\gamma,R}(u) + a_{N,\gamma} R^{-(\gamma-1)} M.$$

Corollary 4.1 (Weighted concentration on the intrinsic scale). *Under the assumptions of Theorem 3.5, let $\lambda > 0$ and $R = \lambda \varepsilon^{1/(\gamma-1)} < \ell$. Then*

$$R^{\gamma-1} \int_0^T \mathcal{D}_{\gamma,R}(u(t)) \, dt \geq \lambda^{\gamma-1} \frac{\ell \bar{m}_\ell(0)}{2} - a_{N,\gamma} T M.$$

Under the standing restriction $R < \ell$, the right-hand side is positive whenever $\lambda^{\gamma-1} \ell \bar{m}_\ell(0)/2 > a_{N,\gamma} T M$.

Proof. Combine the preceding localization inequality with (15), integrate over $[0, T]$, and use $R^{\gamma-1} = \lambda^{\gamma-1}\varepsilon$. $\square$

The weak- $L^{N/(\gamma-1)}$ bound for $|x|^{-(\gamma-1)}$, Lorentz–Hölder, and L^1 - L^p interpolation give the following conversion estimate.

Lemma 4.2 (Weak- L^p conversion estimate). *Let $R > 0$ and $p > N/(N+1-\gamma)$. For every nonnegative $u \in L^1(B_R) \cap L^p(B_R)$,*

$$(16) \quad R^{\gamma-1} \mathcal{D}_{\gamma,R}(u) \leq c_{N,\gamma,p}^{\text{conv}} \left(\int_{B_R} u(x) \, dx \right)^{1 - \frac{(\gamma-1)p'}{N}} (R^{N/p'} \|u\|_{L^p(B_R)})^{\frac{(\gamma-1)p'}{N}}.$$

If $R^{N/p'} \|u\|_{L^p(B_R)} > 0$, then

$$(17) \quad \int_{B_R} u(x) \, dx \geq (c_{N,\gamma,p}^{\text{conv}})^{-\frac{N}{N-(\gamma-1)p'}} (R^{N/p'} \|u\|_{L^p(B_R)})^{-\frac{(\gamma-1)p'}{N-(\gamma-1)p'}} (R^{\gamma-1} \mathcal{D}_{\gamma,R}(u))^{\frac{N}{N-(\gamma-1)p'}}.$$

If $R^{N/p'} \|u\|_{L^p(B_R)} = 0$, then both $\int_{B_R} u(x) \, dx$ and $\mathcal{D}_{\gamma,R}(u)$ vanish, and only (16) is used.

Proof. The Lorentz–Hölder inequality and $|x|^{-(\gamma-1)} \in L^{N/(\gamma-1),\infty}(B_R)$ yield the following bound for the weighted diffusion quantity:

$$(18) \quad \mathcal{D}_{\gamma,R}(u) \leq C_{\text{Lor}} \|u\|_{L^{N/(N+1-\gamma),1}(B_R)},$$

where $C_{\text{Lor}} = C_{\text{Lor}}(N, \gamma)$ includes the factor $a_{N,\gamma}$. Real interpolation between $L^1(B_R)$ and $L^p(B_R)$, with interpolation parameter $(\gamma-1)p'/N$, yields

$$\|u\|_{L^{N/(N+1-\gamma),1}(B_R)} \leq c_{N,\gamma,p}^{\text{int}} \|u\|_{L^1(B_R)}^{1 - \frac{(\gamma-1)p'}{N}} \|u\|_{L^p(B_R)}^{\frac{(\gamma-1)p'}{N}}.$$

Combining this with (18), we absorb the Lorentz constant and the L^1 - L^p interpolation constant into $c_{N,\gamma,p}^{\text{conv}}$. Using

$$\gamma - 1 = \frac{N}{p'} \frac{(\gamma-1)p'}{N}$$

we obtain (16). If $R^{N/p'} \|u\|_{L^p(B_R)} > 0$, solving that inequality for $\int_{B_R} u(x) \, dx$ gives (17); if this factor is zero, then $u = 0$ a.e. on B_R , and both $\int_{B_R} u(x) \, dx$ and $\mathcal{D}_{\gamma,R}(u)$ are zero. $\square$

Theorem 4.3 (Time-integrated conversion to small-ball mass concentration). *Let $\ell > 0$, and let u be the nonnegative radial global mild solution of (1) with $K = -|x|$, diffusivity $\varepsilon > 0$, and conserved mass M . Assume $\bar{m}_\ell(0) > 0$ and*

$$T \geq \frac{6\ell}{\bar{m}_\ell(0)} - \frac{2\ell}{M}.$$

Let $\lambda > 0$ and $R = \lambda \varepsilon^{1/(\gamma-1)} < \ell$. Let $p > N/(N+1-\gamma)$ and suppose that

$$\sup_{0 \leq t \leq T} R^{N/p'} \|u(t)\|_{L^p(B_R)} < \infty.$$

If

$$\lambda^{\gamma-1} \frac{\ell \bar{m}_\ell(0)}{2} - a_{N,\gamma} T M > 0,$$

then the supremum in the preceding display is positive, and

$$(19) \quad \begin{aligned} \int_0^T m_R(t) \, dt &\geq (c_{N,\gamma,p}^{\text{conv}})^{-\frac{N}{N-(\gamma-1)p'}} \left(\sup_{0 \leq t \leq T} R^{N/p'} \|u(t)\|_{L^p(B_R)} \right)^{-\frac{(\gamma-1)p'}{N-(\gamma-1)p'}} \\ &\quad \times T^{-\frac{(\gamma-1)p'}{N-(\gamma-1)p'}} \left(\lambda^{\gamma-1} \frac{\ell \bar{m}_\ell(0)}{2} - a_{N,\gamma} T M \right)^{\frac{N}{N-(\gamma-1)p'}}. \end{aligned}$$

Thus the time integral of the mass in $B_{\lambda R_\varepsilon}$ admits a strictly positive lower bound.

Proof. Corollary 4.1 gives

$$\int_0^T R^{\gamma-1} \mathcal{D}_{\gamma,R}(u(t)) \, dt \geq \lambda^{\gamma-1} \frac{\ell \tilde{m}_\ell(0)}{2} - a_{N,\gamma} T M.$$

If the supremum in the statement were zero, then $u(t) = 0$ a.e. on B_R for every $t \in [0, T]$, so the left-hand side above would vanish, contradicting the strict positivity assumption. Hence that supremum is positive. Applying Lemma 4.2 with the L^p factor bounded by its time supremum gives

$$R^{\gamma-1} \mathcal{D}_{\gamma,R}(u(t)) \leq c_{N,\gamma,p}^{\text{conv}} m_R(t)^{1-\frac{(\gamma-1)p'}{N}} \left(\sup_{0 \leq s \leq T} R^{N/p'} \|u(s)\|_{L^p(B_R)} \right)^{\frac{(\gamma-1)p'}{N}}.$$

Integrating in time and using the concavity of $s \mapsto s^{1-\frac{(\gamma-1)p'}{N}}$,

$$\lambda^{\gamma-1} \frac{\ell \tilde{m}_\ell(0)}{2} - a_{N,\gamma} T M \leq c_{N,\gamma,p}^{\text{conv}} \left(T \sup_{0 \leq s \leq T} R^{N/p'} \|u(s)\|_{L^p(B_R)} \right)^{\frac{(\gamma-1)p'}{N}} \left(\int_0^T m_R(t) \, dt \right)^{1-\frac{(\gamma-1)p'}{N}}.$$

Solving this inequality gives (19). $\square$

Remark 4.4. The L^p condition in Theorem 4.3 is consistent with mass-preserving scaling. If

$$\|u(t)\|_{L^p(B_R)} \lesssim M R^{-N/p'} \quad R \sim \left(\frac{\varepsilon}{M} \right)^{1/(\gamma-1)},$$

then the theorem gives a time-integrated small-ball mass lower bound independent of ε . Proposition 4.5 supplies such a global estimate.

4.2. Fractional L^p upper bound. We first work on compact subintervals of the maximal mild existence interval $[0, T_{\max})$ furnished by Proposition A.1. Semigroup duality gives the distributional formulation on such intervals. To justify testing with u^{p-1} , we use Steklov averages in time, smooth convex approximations of $s \mapsto s^p/p$, and spatial cutoffs. The energy inequality below then follows by passing to the limit using the Dirichlet-form formulation, the Stroock–Varopoulos inequality, weak lower semicontinuity, and Fatou’s lemma. The negative-part estimate and the tent-moment identity are regularized separately in Appendix A.

The proof combines three estimates: fractional Stroock–Varopoulos dissipation, a compression estimate based on the Hardy–Littlewood–Sobolev inequality in Lorentz spaces for $N \geq 2$ (or on the one-dimensional Dirac mass when $N = 1$), and a fractional Nash lower bound. Their constants are independent of $T_{\max}$, so the resulting estimate also supplies the continuation step in Appendix A. Before continuation, the argument uses only the local L^1 bound from Proposition A.1, together with the mass identity and nonnegativity established independently in Appendix A.

Proposition 4.5 (Fractional L^p upper bound). *Let $2 \leq p < \infty$, let $u_0 \geq 0$ belong to $L^1(\mathbb{R}^N) \cap L^\infty(\mathbb{R}^N)$, and let u be the maximal mild solution constructed in Proposition A.1. Then, throughout its maximal existence interval $[0, T_{\max})$,*

$$(20) \quad \|u(t)\|_{L^p} \leq \max \left\{ \|u_0\|_{L^p}, c_{N,\gamma,p}^{\text{up}} M \left(\frac{M}{\varepsilon} \right)^{\frac{N(p-1)}{p(\gamma-1)}} \right\}.$$

For every $T < T_{\max}$, the same argument also gives

$$u^{p/2} \in L^2(0, T; \dot{H}^{\gamma/2}(\mathbb{R}^N)).$$

Here $c_{N,\gamma,p}^{\text{up}} > 0$ depends only on N, γ, p . Applied with $p = p_*$, this estimate rules out finite-time breakdown and yields $T_{\max} = \infty$. Consequently (20) holds for every $t \geq 0$. In particular, for the fixed initial datum considered here, as $\varepsilon \rightarrow 0$,

$$\|u(t)\|_{L^p} \lesssim \varepsilon^{-\frac{N(p-1)}{p(\gamma-1)}}.$$

Lemma 4.6 (Compression estimate). *Let $2 \leq p < \infty$, $u \geq 0$, and $\int_{\mathbb{R}^N} u \, dx = M$. For $N \geq 2$,*

$$(21) \quad \int_{\mathbb{R}^N} u(x)^p (|\cdot|^{-1} * u)(x) \, dx \leq c_{N,\gamma,p}^{\text{comp}} M \|u\|_{L^p}^{p(1-1/\gamma)} \|\Lambda^{\gamma/2}(u^{p/2})\|_{L^2}^{2/\gamma}.$$

For $N = 1$,

$$(22) \quad \int_{\mathbb{R}} u^{p+1} dx \leq c_{\gamma,p}^{\text{comp}} M \|u\|_{L^p}^{p(1-1/\gamma)} \|\Lambda^{\gamma/2}(u^{p/2})\|_{L^2}^{2/\gamma}.$$

Proof. Assume first that $N \geq 2$. By the endpoint Hardy–Littlewood–Sobolev inequality in Lorentz form [17, 19], with constant $C_{\text{HLS}} = C_{\text{HLS}}(N)$,

$$\| |\cdot|^{-1} * u \|_{L^{N,\infty}} \leq C_{\text{HLS}} \|u\|_{L^1} = C_{\text{HLS}} M.$$

Using Lorentz–Hölder,

$$\int_{\mathbb{R}^N} u(x)^p (|\cdot|^{-1} * u)(x) dx \leq C_{\text{HLS}} M \|u^p\|_{L^{N/(N-1),1}}.$$

By Lorentz interpolation,

$$\|u^p\|_{L^{N/(N-1),1}} \leq c_{N,\gamma,p}^{\text{int}} \|u\|_{L^p}^{p(1-1/\gamma)} \|u\|_{L^{pN/(N-\gamma)}}^{p/\gamma}.$$

The fractional Sobolev inequality [27], with constant $C_{\text{Sob}} = C_{\text{Sob}}(N, \gamma)$, gives

$$\|u\|_{L^{pN/(N-\gamma)}}^p = \|u^{p/2}\|_{L^{2N/(N-\gamma)}}^2 \leq C_{\text{Sob}} \left\| \Lambda^{\gamma/2}(u^{p/2}) \right\|_{L^2}^2.$$

This proves (21), after grouping the Hardy–Littlewood–Sobolev, Lorentz-interpolation, and fractional Sobolev constants into $c_{N,\gamma,p}^{\text{comp}}$.

When $N = 1$, $K'' = -2\delta_0$, so the compression term is local. We estimate it by a one-dimensional fractional interpolation inequality:

$$\int_{\mathbb{R}} u^{p+1} dx = \int_{\mathbb{R}} u (u^{p/2})^2 dx \leq M \|u^{p/2}\|_{L^\infty}^2.$$

Since $\gamma/2 > 1/2$, the one-dimensional fractional Gagliardo–Nirenberg estimate [27], with constant $c_{\text{GN}} = c_{\text{GN}}(\gamma, p)$, gives

$$\|u^{p/2}\|_{L^\infty}^2 \leq c_{\text{GN}} \|\Lambda^{\gamma/2}(u^{p/2})\|_{L^2}^{2/\gamma} \|u^{p/2}\|_{L^2}^{2(1-1/\gamma)} = c_{\text{GN}} \left\| \Lambda^{\gamma/2}(u^{p/2}) \right\|_{L^2}^{2/\gamma} \|u\|_{L^p}^{p(1-1/\gamma)}.$$

This proves (22), after grouping the one-dimensional Gagliardo–Nirenberg constant into the compression constant in (22). $\square$

Lemma 4.7 (Fractional Nash lower bound). *Let $2 \leq p < \infty$, $u \geq 0$, and $\int_{\mathbb{R}^N} u dx = M$. Then*

$$(23) \quad \left\| \Lambda^{\gamma/2}(u^{p/2}) \right\|_{L^2}^2 \geq c_{N,\gamma,p}^{\text{Nash}} M^{-\frac{p\gamma}{N(p-1)}} (\|u\|_{L^p}^p)^{1+\frac{\gamma}{N(p-1)}}.$$

Proof. The fractional Nash inequality [14, 18, 27], with constant $C_{\text{Nash}} = C_{\text{Nash}}(N, \gamma)$, applied to $u^{p/2}$ gives

$$\|u^{p/2}\|_{L^2}^{2+2\gamma/N} \leq C_{\text{Nash}} \|u^{p/2}\|_{L^1}^{2\gamma/N} \|\Lambda^{\gamma/2}(u^{p/2})\|_{L^2}^2.$$

Since $p \geq 2$, Hölder interpolation and mass conservation imply

$$\|u^{p/2}\|_{L^1} = \int_{\mathbb{R}^N} u^{p/2} dx \leq M^{\frac{p}{2(p-1)}} \|u\|_{L^p}^{\frac{p(p-2)}{2(p-1)}}.$$

Substitute this estimate into the Nash inequality and use $\|u^{p/2}\|_{L^2}^2 = \|u\|_{L^p}^p$. After absorbing the resulting constants into $c_{N,\gamma,p}^{\text{Nash}}$, we obtain (23). $\square$

Proof of Proposition 4.5. Fix $0 < \tau < T < T_{\text{max}}$. Interpolating the local L^∞ bound with continuity in $L^1 \cap L^{p^*}$ gives continuity in every finite L^p space. We record the approximation leading to the energy inequality. Let $\chi_R \in C_c^\infty(\mathbb{R}^N)$ satisfy $\chi_R = 1$ on B_R , $\chi_R = 0$ outside B_{2R} , and $|\nabla \chi_R| \leq C/R$. Mollify the distributional equation in space, take a Steklov average in time, and test it with χ_R times the derivative of a smooth convex approximation of $s \mapsto (s_+)^p/p$. On $[\tau, T]$, the local $L^1 \cap L^\infty$ bounds imply

$$\|\nabla(\nabla K * u(t))\|_{L^\infty} \leq C_N (\|u(t)\|_{L^1} + \|u(t)\|_{L^\infty})$$

when $N \geq 2$; in one dimension, $(K' * u)' = -2u$. Hence the standard Friedrichs commutator for the transport term tends to zero with the spatial mollification parameter. For the fractional term, first truncate the Dirichlet kernel to $|x - y| > \rho$. At this level, all pairings are integrable, and symmetry together with convexity gives the truncated Stroock–Varopoulos inequality.

We let the spatial mollification, Steklov, and convex regularization parameters tend to zero at fixed R and ρ . The only additional term produced by the transport cutoff is bounded by

$$\int_{\tau}^T \left| \frac{1}{p} \int_{\mathbb{R}^N} u(x, t)^p \nabla \chi_R \cdot (\nabla K * u(t))(x) dx \right| dt \leq \frac{CM}{R} \int_{\tau}^T \int_{B_{2R} \setminus B_R} u(x, t)^p dx dt$$

because $\|\nabla K * u(t)\|_{L^\infty} \leq M$. The right-hand side is bounded by $CM(T - \tau) \sup_{\tau < t < T} \|u(t)\|_{L^p}^p / R$. Since u is bounded in $L^\infty(\tau, T; L^p)$, this bound tends to zero as $R \rightarrow \infty$. For fixed ρ , the cutoff terms in the truncated Dirichlet form also vanish by dominated convergence.

Finally, letting $\rho \downarrow 0$, monotone convergence for the nonnegative dissipation and lower semicontinuity of the full Dirichlet form give

$$\begin{aligned} & \frac{1}{p} \|u(T)\|_{L^p}^p + c_p^{\text{SV}} \varepsilon \int_{\tau}^T \left\| \Lambda^{\gamma/2}(u^{p/2})(t) \right\|_{L^2}^2 dt \\ & \leq \frac{1}{p} \|u(\tau)\|_{L^p}^p - \frac{p-1}{p} \int_{\tau}^T \int_{\mathbb{R}^N} u(x, t)^p (\Delta K * u(t))(x) dx dt, \end{aligned}$$

where $c_p^{\text{SV}} := 4(p-1)/p^2$. The diffusion term is controlled by the Stroock–Varopoulos inequality [15, 16]: for almost every $t \in (\tau, T)$,

$$\int_{\mathbb{R}^N} u(t)^{p-1} \Lambda^\gamma u(t) dx \geq c_p^{\text{SV}} \left\| \Lambda^{\gamma/2}(u^{p/2})(t) \right\|_{L^2}^2.$$

For almost every $t \in (\tau, T)$, the aggregation term satisfies

$$\int_{\mathbb{R}^N} u(t)^{p-1} \nabla \cdot (u(t)(\nabla K * u(t))) dx = \frac{p-1}{p} \int_{\mathbb{R}^N} u(t)^p (\Delta K * u(t)) dx.$$

When $N \geq 2$, $\Delta K = -(N-1)|x|^{-1}$ in the distributional sense; when $N = 1$, $K'' = -2\delta_0$. By Lemma 4.6,

$$(24) \quad \frac{1}{p} \frac{d}{dt} \|u(t)\|_{L^p}^p + c_p^{\text{SV}} \varepsilon \left\| \Lambda^{\gamma/2}(u^{p/2})(t) \right\|_{L^2}^2 \leq c_{N,\gamma,p}^{\text{comp}} M \|u(t)\|_{L^p}^{p(1-1/\gamma)} \left\| \Lambda^{\gamma/2}(u^{p/2})(t) \right\|_{L^2}^{2/\gamma}.$$

Young's inequality gives

$$\begin{aligned} c_{N,\gamma,p}^{\text{comp}} M \|u(t)\|_{L^p}^{p(1-1/\gamma)} \left\| \Lambda^{\gamma/2}(u^{p/2})(t) \right\|_{L^2}^{2/\gamma} & \leq \frac{c_p^{\text{SV}} \varepsilon}{2} \left\| \Lambda^{\gamma/2}(u^{p/2})(t) \right\|_{L^2}^2 \\ & \quad + c_{N,\gamma,p}^{\text{Young}} \varepsilon^{-1/(\gamma-1)} M^{\gamma/(\gamma-1)} \|u(t)\|_{L^p}^p. \end{aligned}$$

Substituting this into (24), we obtain

$$(25) \quad \frac{d}{dt} \|u(t)\|_{L^p}^p + c_{N,\gamma,p}^{\text{diss}} \varepsilon \left\| \Lambda^{\gamma/2}(u^{p/2})(t) \right\|_{L^2}^2 \leq c_{N,\gamma,p}^{\text{Young}} \varepsilon^{-1/(\gamma-1)} M^{\gamma/(\gamma-1)} \|u(t)\|_{L^p}^p.$$

Using Lemma 4.7 in (25), we get

$$(26) \quad \frac{d}{dt} \|u(t)\|_{L^p}^p \leq c_{N,\gamma,p}^{\text{Young}} \varepsilon^{-1/(\gamma-1)} M^{\gamma/(\gamma-1)} \|u(t)\|_{L^p}^p - c_{N,\gamma,p}^{\text{diss}} \varepsilon M^{-\frac{p\gamma}{N(p-1)}} (\|u(t)\|_{L^p}^p)^{1+\frac{\gamma}{N(p-1)}}.$$

Whenever

$$\|u(t)\|_{L^p}^p \geq C_{N,\gamma,p}^{\text{thr}} M^{p+\frac{N(p-1)}{\gamma-1}} \varepsilon^{-\frac{N(p-1)}{\gamma-1}},$$

the right-hand side of (26) is nonpositive. By the comparison principle,

$$\|u(t)\|_{L^p}^p \leq \max \left\{ \|u_0\|_{L^p}^p, C_{N,\gamma,p}^{\text{thr}} M^{p+\frac{N(p-1)}{\gamma-1}} \varepsilon^{-\frac{N(p-1)}{\gamma-1}} \right\}.$$

The estimates above are independent of τ . Continuity of u in L^p at $t = 0$, together with Fatou's lemma and lower semicontinuity of the fractional Dirichlet integral, permits $\tau \downarrow 0$. Taking the p -th root gives (20) on $[0, T_{\max})$, and also

$$u^{p/2} \in L^2(0, T; \dot{H}^{\gamma/2}(\mathbb{R}^N)) \quad \text{for every } T < T_{\max}.$$

Choosing $p = p_*$ and combining this uniform bound with conservation of the L^1 norm, the continuation criterion in Proposition A.1 yields $T_{\max} = \infty$. The estimate therefore holds for all $t \geq 0$. $\square$

Remark 4.8 (Difference between the one-dimensional fractional and classical diffusion cases). The mild well-posedness construction in Appendix A applies in every dimension $N \geq 1$; in one dimension radial symmetry means evenness. In the classical case $\gamma = 2$, the distributional identity

$$\frac{d^2}{dx^2}|x| = 2\delta_0$$

shows that the corresponding concentration quantity is the endpoint value $u(0, t)$. The proof in [12] therefore uses an additional H^1 bound to control this point value.

For $1 < \gamma < 2$, the fractional diffusion of the inner-mass tent produces the locally integrable weight $|x|^{-(\gamma-1)} \in L^1_{\text{loc}}(\mathbb{R})$. Thus Lemma 4.2 applies when $N = 1$. The L^p -energy estimate still requires a separate one-dimensional calculation: $K'' = -2\delta_0$, so the aggregation term becomes $\int_{\mathbb{R}} u^{p+1}$, which is controlled by a one-dimensional fractional Gagliardo–Nirenberg inequality.

4.3. Proof of the concentration theorem.

Proof of Theorem 2.3. Let

$$R = \lambda \varepsilon^{1/(\gamma-1)}.$$

Choose $\varepsilon_* > 0$ so that the scale restriction $R < \ell$ and the domination of the fixed initial L^p norm by the ε -dependent term in (20) both hold. Explicitly, it is enough to impose

$$\varepsilon_* \leq \min \left\{ \left(\frac{\ell}{\lambda} \right)^{\gamma-1}, M \left(\frac{c_{N,\gamma,p}^{\text{up}} M}{\|u_0\|_{L^p}} \right)^{\frac{p(\gamma-1)}{N(p-1)}} \right\}.$$

Fix $0 < \varepsilon < \varepsilon_*$, and write $u = u^\varepsilon$. By the choice of ε_* , the ε -dependent term in (20) dominates the fixed initial norm. Proposition 4.5 therefore gives

$$\|u(t)\|_{L^p} \leq c_{N,\gamma,p}^{\text{up}} M \left(\frac{M}{\varepsilon} \right)^{\frac{N(p-1)}{p(\gamma-1)}}, \quad 0 < \varepsilon < \varepsilon_*, \quad t \geq 0.$$

It follows that

$$R^{N/p'} \|u(t)\|_{L^p(B_R)} \leq c_{N,\gamma,p}^{\text{up}} \lambda^{N/p'} M^{1+\frac{N}{p'(\gamma-1)}},$$

where the right-hand side is independent of ε . Substituting this bound into Theorem 4.3, we obtain

$$\begin{aligned} \int_0^T m_R(t) dt &\geq (c_{N,\gamma,p}^{\text{conv}})^{-\frac{N}{N-(\gamma-1)p'}} \left(c_{N,\gamma,p}^{\text{up}} \lambda^{N/p'} M^{1+\frac{N}{p'(\gamma-1)}} \right)^{-\frac{(\gamma-1)p'}{N-(\gamma-1)p'}} \\ &\quad \times T^{-\frac{(\gamma-1)p'}{N-(\gamma-1)p'}} \left(\lambda^{\gamma-1} \frac{\ell \tilde{m}_\ell(0)}{2} - a_{N,\gamma} T M \right)^{\frac{N}{N-(\gamma-1)p'}}. \end{aligned}$$

Since the last factor is positive by assumption, define $C_* > 0$ to be the right-hand side. This is precisely (3). $\square$

Proof of Corollary 2.4. If the conclusion were false, then

$$\int_{B_{\lambda \varepsilon^{1/(\gamma-1)}}} u^\varepsilon(x, t) dx < \frac{C_*}{T} \quad \text{for every } t \in [0, T].$$

Integrating this strict inequality over $[0, T]$ would contradict (3). Hence a time $t_\varepsilon \in [0, T]$ satisfying (4) must exist. $\square$

Proof of Corollary 2.6. Let $R = \lambda \varepsilon^{1/(\gamma-1)}$ and write $u = u^\varepsilon$. By Hölder's inequality,

$$m_R(t) = \int_{B_R} u(x, t) dx \leq |B_R|^{1-\frac{1}{q}} \left(\int_{B_R} u(x, t)^q dx \right)^{1/q}.$$

Integrating in time over $[0, T]$ and using Theorem 2.3, we obtain

$$\int_0^T \left(\int_{B_R} u(x, t)^q dx \right)^{1/q} dt \geq |B_R|^{-(1-\frac{1}{q})} C_*.$$

Since $|B_R| = |B_1| \lambda^N \varepsilon^{N/(\gamma-1)}$, this becomes

$$\int_0^T \left(\int_{B_{\lambda \varepsilon^{1/(\gamma-1)}}} u(x, t)^q dx \right)^{1/q} dt \geq C_* |B_1|^{-\frac{q-1}{q}} \lambda^{-\frac{N(q-1)}{q}} \varepsilon^{-\frac{N(q-1)}{q(\gamma-1)}}.$$

This is (5). □

APPENDIX A. GLOBAL WELL-POSEDNESS AND ADMISSIBILITY OF THE TENT MOMENT

A.1. Mild well-posedness and structural properties. This appendix proves Theorem 2.1. The construction applies to general initial data, and uniqueness preserves radial symmetry when the initial datum is radial. The bounded cusp force satisfies

$$(27) \quad \|\nabla K * u\|_{L^\infty} \leq \|u\|_{L^1}, \quad \|u(\nabla K * v)\|_{L^q} \leq \|u\|_{L^q} \|v\|_{L^1}, \quad 1 \leq q \leq \infty.$$

We also use the fractional heat-semigroup estimates

$$(28) \quad \|e^{-\varepsilon t \Lambda^\gamma} f\|_{L^r} \leq C(\varepsilon t)^{-\frac{N}{\gamma}(\frac{1}{q}-\frac{1}{r})} \|f\|_{L^q}, \quad \|\nabla e^{-\varepsilon t \Lambda^\gamma} f\|_{L^r} \leq C(\varepsilon t)^{-\frac{1}{\gamma}-\frac{N}{\gamma}(\frac{1}{q}-\frac{1}{r})} \|f\|_{L^q},$$

for $1 \leq q \leq r \leq \infty$. In particular, $\int_0^t (t-s)^{-1/\gamma} ds < \infty$ because $\gamma > 1$.

Fix $p_* > \max\{2, N/(\gamma-1)\}$.

Proposition A.1 (Local mild well-posedness and continuation criterion). *For every $u_0 \in L^1(\mathbb{R}^N) \cap L^{p_*}(\mathbb{R}^N) \cap L^\infty(\mathbb{R}^N)$, there exists a maximal time $T_{\max} \in (0, \infty]$ and a mild solution*

$$u \in C([0, T_{\max}); L^1(\mathbb{R}^N) \cap L^{p_*}(\mathbb{R}^N)) \cap L_{\text{loc}}^\infty([0, T_{\max}); L^\infty(\mathbb{R}^N))$$

satisfying (2). This solution is unique within the displayed class. If $T_{\max} < \infty$, then

$$(29) \quad \limsup_{t \uparrow T_{\max}} (\|u(t)\|_{L^1} + \|u(t)\|_{L^{p_*}}) = \infty.$$

Proof. For $T > 0$, let

$$X_T := C([0, T]; L^1(\mathbb{R}^N) \cap L^{p_*}(\mathbb{R}^N)) \cap L^\infty(0, T; L^\infty(\mathbb{R}^N)),$$

endowed with

$$\|u\|_{X_T} := \sup_{0 \leq t \leq T} (\|u(t)\|_{L^1} + \|u(t)\|_{L^{p_*}} + \|u(t)\|_{L^\infty}).$$

Consider the right-hand side of (2) as a map on X_T . For $q \in \{1, p_*, \infty\}$, (27) and (28) give

$$\begin{aligned} & \left\| e^{-\varepsilon t \Lambda^\gamma} u_0 - \int_0^t \nabla \cdot e^{-\varepsilon(t-s)\Lambda^\gamma} [u(s)(\nabla K * u(s))] ds \right\|_{L^q} \\ & \leq \|u_0\|_{L^q} + C\varepsilon^{-1/\gamma} \int_0^t (t-s)^{-1/\gamma} \|u(s)\|_{L^q} \|u(s)\|_{L^1} ds. \end{aligned}$$

Consequently,

$$\begin{aligned} & \sup_{0 \leq t \leq T} (\|u(t)\|_{L^1} + \|u(t)\|_{L^{p_*}} + \|u(t)\|_{L^\infty}) \leq \\ & C(\|u_0\|_{L^1} + \|u_0\|_{L^{p_*}} + \|u_0\|_{L^\infty}) + C\varepsilon^{-1/\gamma} T^{1-1/\gamma} \left[\sup_{0 \leq t \leq T} (\|u(t)\|_{L^1} + \|u(t)\|_{L^{p_*}} + \|u(t)\|_{L^\infty}) \right]^2. \end{aligned}$$

For two functions u and v , the identity $u(\nabla K * u) - v(\nabla K * v) = (u-v)(\nabla K * u) + v(\nabla K * (u-v))$ gives the corresponding Lipschitz estimate with the factor

$$C\varepsilon^{-1/\gamma} T^{1-1/\gamma} \sup_{0 \leq t \leq T} (\|u(t)\|_{L^1} + \|u(t)\|_{L^{p_*}} + \|u(t)\|_{L^\infty} + \|v(t)\|_{L^1} + \|v(t)\|_{L^{p_*}} + \|v(t)\|_{L^\infty}).$$

Let $R_0 := 2C(\|u_0\|_{L^1} + \|u_0\|_{L^{p_*}} + \|u_0\|_{L^\infty})$. Choose $T > 0$ so that $C\varepsilon^{-1/\gamma} T^{1-1/\gamma} R_0 \leq 1/4$. Then the mild map sends the closed X_T -ball of radius R_0 into itself and is a contraction there. This gives a unique local solution. The L^1 and L^{p_*} components provide strong continuity at $t = 0$, whereas the L^∞ component supplies the local bound.

Iterating the local construction yields a maximal solution. Assume that $T_{\max} < \infty$ and $\sup_{t < T_{\max}} (\|u(t)\|_{L^1} + \|u(t)\|_{L^{p_*}}) < \infty$. Choose $h \in (0, T_{\max}/2)$. Restarting (2) at $t - h$ and applying (28) with $q = p_*$ and $r = \infty$, we obtain for $t \geq h$

$$\|u(t)\|_{L^\infty} \leq C(\varepsilon h)^{-\frac{N}{\gamma p_*}} \|u(t-h)\|_{L^{p_*}} + C \sup_{s < T_{\max}} \|u(s)\|_{L^1} \int_{t-h}^t (\varepsilon(t-s))^{-\frac{1}{\gamma} - \frac{N}{\gamma p_*}} \|u(s)\|_{L^{p_*}} ds.$$

The condition $p_* > N/(\gamma - 1)$ makes the time integral finite. Hence u is uniformly bounded in L^∞ near $T_{\max}$. The local construction can therefore be restarted with a lifespan bounded below uniformly, contradicting maximality. This proves (29). $\square$

On every compact subinterval of $(0, T_{\max})$, semigroup duality yields the distributional formulation for the local mild solution. The negative-part calculation in Proposition A.3 is justified by Steklov averaging in time and smooth convex approximation. Conservation of mass is handled separately by spatial cutoffs, while Proposition A.5 uses smooth compactly supported approximations of the tent weight.

Conservation of mass and nonnegativity.

Lemma A.2 (Conservation of the spatial integral). *For every $t < T_{\max}$,*

$$\int_{\mathbb{R}^N} u(x, t) dx = \int_{\mathbb{R}^N} u_0(x) dx = M.$$

Proof. Fix $T < T_{\max}$. The local mild construction already gives

$$\sup_{0 \leq s \leq T} \|u(s)\|_{L^1} < \infty.$$

Hence

$$\|u(s)(\nabla K * u(s))\|_{L^1} \leq \left(\sup_{0 \leq r \leq T} \|u(r)\|_{L^1} \right)^2$$

for $0 \leq s \leq T$, because $|\nabla K| \leq 1$. Integrating (2) over $\mathbb{R}^N$, the fractional heat semigroup preserves the integral. To justify the Duhamel term on the whole space, choose $\chi_R(x) = \chi(x/R)$, where $\chi \in C_c^\infty(\mathbb{R}^N)$, $\chi \equiv 1$ on B_1 , and $|\nabla \chi_R| \leq C/R$. Pairing the Duhamel term with χ_R and moving the semigroup onto the test function gives, for $0 \leq t \leq T$, an error bounded by

$$\int_0^t \frac{C}{R} \|u(s)(\nabla K * u(s))\|_{L^1} ds \leq \frac{Ct}{R} \left(\sup_{0 \leq r \leq T} \|u(r)\|_{L^1} \right)^2,$$

which tends to zero as $R \rightarrow \infty$. Letting $R \rightarrow \infty$ gives conservation of the spatial integral on $[0, T]$. Since $T < T_{\max}$ was arbitrary, the assertion follows. $\square$

Proposition A.3 (Preservation of nonnegativity). *If $u_0 \geq 0$, then $u(t) \geq 0$ for every $t < T_{\max}$. Consequently, $\|u(t)\|_{L^1} = M$.*

Proof. Let $u_- := \max\{-u, 0\}$. Since $p_* > 2$, interpolation gives $u \in C([0, T]; L^2(\mathbb{R}^N))$ for every $T < T_{\max}$. The following calculation is justified by spatial mollification, Steklov averaging in time, and a smooth convex approximation of $s \mapsto \frac{1}{2}(s_-)^2$. The transport commutator vanishes by the same local Lipschitz bound for $\nabla K * u$ used in the proof of Proposition 4.5. For the fractional term, use the truncated representation

$$\Lambda^\gamma u(x) = c_{N, \gamma} \lim_{\rho \downarrow 0} \int_{\mathbb{R}^N \setminus B_\rho(x)} \frac{u(x) - u(y)}{|x - y|^{N+\gamma}} dy.$$

For fixed ρ , symmetry of the set $\{(x, y) \in \mathbb{R}^N \times \mathbb{R}^N : |x - y| > \rho\}$ gives

$$\begin{aligned} & -c_{N, \gamma} \int_{\mathbb{R}^N} u_-(x) \int_{\mathbb{R}^N \setminus B_\rho(x)} \frac{u(x) - u(y)}{|x - y|^{N+\gamma}} dy dx \\ &= -\frac{c_{N, \gamma}}{2} \iint_{\substack{x, y \in \mathbb{R}^N \\ |x - y| > \rho}} \frac{(u_-(x) - u_-(y))(u(x) - u(y))}{|x - y|^{N+\gamma}} dx dy. \end{aligned}$$

This identity is obtained by symmetrizing in x and y . For all $a, b \in \mathbb{R}$, $(a - b)(a - b_-) \leq -|a - b_-|^2$. Consequently,

$$-c_{N,\gamma} \int_{\mathbb{R}^N} u_-(x) \int_{\mathbb{R}^N \setminus B_\rho(x)} \frac{u(x) - u(y)}{|x - y|^{N+\gamma}} dy dx \geq \frac{c_{N,\gamma}}{2} \iint_{\substack{x, y \in \mathbb{R}^N \\ |x - y| > \rho}} \frac{|u_-(x) - u_-(y)|^2}{|x - y|^{N+\gamma}} dx dy.$$

Passing to the limits in the kernel truncation and the smooth convex approximation, and using monotone convergence on the nonnegative right-hand side, yields

$$- \int_{\mathbb{R}^N} u_- \Lambda^\gamma u dx \geq \frac{c_{N,\gamma}}{2} \iint_{\mathbb{R}^N \times \mathbb{R}^N} \frac{|u_-(x) - u_-(y)|^2}{|x - y|^{N+\gamma}} dx dy = \|\Lambda^{\gamma/2} u_-\|_{L^2}^2.$$

The transport term satisfies

$$\int_{\mathbb{R}^N} \nabla \cdot (u(\nabla K * u))(-u_-) dx = \frac{1}{2} \int_{\mathbb{R}^N} (\Delta K * u) u_-^2 dx.$$

Therefore, for almost every $t \in (0, T)$,

$$(30) \quad \frac{1}{2} \frac{d}{dt} \|u_-(t)\|_{L^2}^2 + \varepsilon \|\Lambda^{\gamma/2} u_-(t)\|_{L^2}^2 \leq \frac{1}{2} \|\Delta K * u(t)\|_{L^\infty} \|u_-(t)\|_{L^2}^2.$$

For $N \geq 2$, $\Delta K * u = -(N - 1)|\cdot|^{-1} * u$. For every $x \in \mathbb{R}^N$, split the convolution at unit distance:

$$\begin{aligned} |(|\cdot|^{-1} * u)(x)| &\leq \int_{|x-y|<1} \frac{|u(y)|}{|x-y|} dy + \int_{|x-y|\geq 1} \frac{|u(y)|}{|x-y|} dy \\ &\leq \|u\|_{L^\infty} \int_{|z|<1} \frac{dz}{|z|} + \|u\|_{L^1} \\ &= \frac{\sigma_N}{N-1} \|u\|_{L^\infty} + \|u\|_{L^1}. \end{aligned}$$

The near-field estimate follows after setting $z = x - y$, while the far-field estimate uses $|x - y|^{-1} \leq 1$ on $\{|x - y| \geq 1\}$. Therefore

$$\|\Delta K * u\|_{L^\infty} \leq \sigma_N \|u\|_{L^\infty} + (N - 1) \|u\|_{L^1}.$$

For $N = 1$, $K'' * u = -2u$ and $\|K'' * u\|_{L^\infty} \leq 2\|u\|_{L^\infty}$. By Proposition A.1, the coefficient in (30) belongs to $L^1(0, T)$ for every $T < T_{\max}$. Since $u_-(0) = 0$, Gronwall's inequality gives $u_- \equiv 0$. $\square$

Global continuation and radial symmetry.

Proof of the global-existence part of Theorem 2.1. Applying Proposition 4.5 with $p = p_*$ gives $\sup_{0 \leq t < T_{\max}} \|u(t)\|_{L^{p_*}} < \infty$. Proposition A.3 gives $\|u(t)\|_{L^1} = M$. The continuation criterion (29) now yields $T_{\max} = \infty$. The restart estimate in the proof of Proposition A.1 yields a uniform L^∞ bound on $[\tau, \infty)$ for every $\tau > 0$. $\square$

Corollary A.4 (Preservation of radial symmetry). *If u_0 is radial, then the global mild solution is radial for every $t \geq 0$.*

Proof. For every $Q \in O(N)$, the function $(x, t) \mapsto u(Qx, t)$ satisfies the same mild equation and has the same initial datum. The fractional heat semigroup is rotation invariant, and $\nabla K(Qx) = Q \nabla K(x)$ for a.e. $x \in \mathbb{R}^N$. Uniqueness gives $u(Qx, t) = u(x, t)$. In dimension one this identity expresses preservation of evenness. $\square$

A.2. Weak formulation and passage to the tent weight. The global mild solution is also a distributional solution: for every $\psi \in C_c^\infty(\mathbb{R}^N \times [0, \infty))$,

$$\int_0^\infty \int_{\mathbb{R}^N} (-u \partial_t \psi + \varepsilon u \Lambda^\gamma \psi - u(\nabla K * u) \cdot \nabla \psi) dx dt = \int_{\mathbb{R}^N} u_0(x) \psi(x, 0) dx.$$

The standard semigroup duality argument applied to (2) proves this identity. All terms are integrable; in particular, the transport term satisfies $\|u(\nabla K * u)\|_{L^1} \leq M^2$.

Proposition A.5 (Admissibility of the tent weight). *For every $\ell > 0$,*

$$\bar{m}_\ell(t) = \int_{\mathbb{R}^N} \left(1 - \frac{|x|}{\ell}\right)_+ u(x, t) \, dx \in AC_{\text{loc}}([0, \infty)).$$

Moreover, for every $0 \leq t_1 \leq t_2 < \infty$, the following identity holds, with the fractional term understood through the Dirichlet pairing; for almost every $t > 0$, Proposition 3.1 identifies this pairing with the displayed integral:

$$(31) \quad \begin{aligned} \bar{m}_\ell(t_2) - \bar{m}_\ell(t_1) = & -\varepsilon \int_{t_1}^{t_2} \int_{\mathbb{R}^N} u(x, t) \Lambda^\gamma \left(1 - \frac{|x|}{\ell}\right)_+ \, dx \, dt \\ & + \int_{t_1}^{t_2} \int_{\mathbb{R}^N} u(x, t) (\nabla K * u(t))(x) \cdot \nabla \left(1 - \frac{|x|}{\ell}\right)_+ \, dx \, dt. \end{aligned}$$

Proof. With ϕ_ℓ denoting the tent weight, let ρ_n be a standard mollifier and set $\phi_{\ell,n} = \rho_n * \phi_\ell$. They belong to $C_c^\infty(\mathbb{R}^N)$, converge to ϕ_ℓ in $H^{\gamma/2}(\mathbb{R}^N)$ and uniformly on $\mathbb{R}^N$, and satisfy

$$\sup_n \|\nabla \phi_{\ell,n}\|_{L^\infty} \leq \frac{1}{\ell}, \quad \nabla \phi_{\ell,n}(x) \rightarrow \nabla \phi_\ell(x) \quad \text{for a.e. } x.$$

Apply the weak formulation with $\psi(x, t) = \eta(t) \phi_{\ell,n}(x)$, where $\eta \in C_c^\infty((0, \infty))$ is arbitrary. For the transport term, the bound $|\nabla K * u(x, t)| \leq M$ and dominated convergence give

$$\int_0^\infty \eta(t) \int_{\mathbb{R}^N} u(\nabla K * u) \cdot \nabla \phi_{\ell,n} \, dx \, dt \longrightarrow \int_0^\infty \eta(t) \int_{\mathbb{R}^N} u(\nabla K * u) \cdot \nabla \phi_\ell \, dx \, dt.$$

Indeed, the integrands are bounded in absolute value by $(M/\ell)|u(x, t)||\eta(t)|$, which is integrable in space and time on the support of η .

For the fractional term, Proposition 4.5 with $p = 2$ gives $\Lambda^{\gamma/2} u \in L^2(0, T; L^2(\mathbb{R}^N))$ for every $T > 0$. Using the fractional Dirichlet pairing,

$$\int_{\mathbb{R}^N} u \Lambda^\gamma (\phi_{\ell,n} - \phi_\ell) \, dx = \int_{\mathbb{R}^N} \Lambda^{\gamma/2} u \Lambda^{\gamma/2} (\phi_{\ell,n} - \phi_\ell) \, dx,$$

we obtain, for every finite $T > 0$,

$$\int_0^T \left| \int_{\mathbb{R}^N} u \Lambda^\gamma (\phi_{\ell,n} - \phi_\ell) \, dx \right| \, dt \leq T^{1/2} \|\Lambda^{\gamma/2} u\|_{L^2(0, T; L^2)} \|\phi_{\ell,n} - \phi_\ell\|_{H^{\gamma/2}} \longrightarrow 0.$$

For almost every $t > 0$, we also have $u(t) \in H^{\gamma/2}(\mathbb{R}^N) \cap L^1(\mathbb{R}^N) \cap L^\infty(\mathbb{R}^N)$. To identify the Dirichlet pairing with the transposition pairing in Proposition 3.1, approximate $u(t)$ by nonnegative smooth compactly supported functions u_j obtained by truncation and mollification. These functions may be chosen so that $u_j \rightarrow u(t)$ in $H^{\gamma/2} \cap L^1$ and almost everywhere, with $\sup_j \|u_j\|_{L^\infty} \leq \|u(t)\|_{L^\infty}$. The two pairings agree for u_j . Their Dirichlet forms converge by the $H^{\gamma/2}$ convergence, while the absolutely convergent decomposition and the bounds in the proof of Proposition 3.1 give convergence of the transposition integrals. Thus the pairings agree for $u(t)$.

Thus the weak formulation passes to the limit with the tent weight. The same estimates show that

$$t \longmapsto -\varepsilon \langle \Lambda^{\gamma/2} u(t), \Lambda^{\gamma/2} \phi_\ell \rangle + \int_{\mathbb{R}^N} u(x, t) (\nabla K * u(t))(x) \cdot \nabla \phi_\ell(x) \, dx$$

belongs to $L^1(0, T)$ for every $T > 0$. Hence $t \mapsto \bar{m}_\ell(t)$ has the displayed function as its distributional derivative and admits an absolutely continuous representative on $[0, T]$. Continuity of u in L^1 identifies its value at $t = 0$.

Consequently, (31) holds for every $0 \leq t_1 \leq t_2 < \infty$, with the fractional term understood through the Dirichlet pairing. Proposition 3.1 identifies this pairing with the displayed integral for almost every positive time. $\square$

Combining Propositions A.5, 3.1, and 3.3 yields Proposition 3.4.

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DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

ETHICS APPROVAL

Not applicable.

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SCHOOL OF MATHEMATICAL SCIENCES, BEIHANG UNIVERSITY, BEIJING 100191
Email address: ruiyizhang@buaa.edu.cn

SCHOOL OF MATHEMATICAL SCIENCES, BEIHANG UNIVERSITY, BEIJING 100191
Email address: xiaoxinzheng@buaa.edu.cn